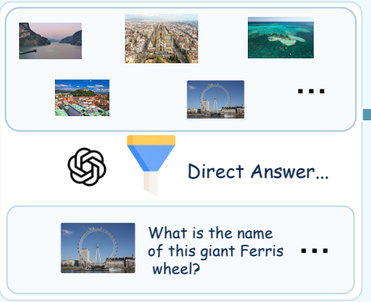
A Controllable Data Refinement: *From Easy VQA to Hard Long-Horizon Multimodal Search Tasks*

1 Seed Collection

Easy multimodal VQA seeds: LiveVQA, FVQA, ...



Seed VQA dataset D_0

2 Refine: Make It Harder via Controlled Transformations

2.1 BuildLink

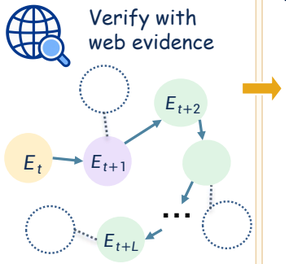
Convert to an explicit entity chain



Entity Chain C^0

2.2 Expand (L)

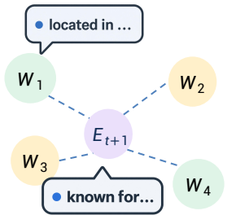
Verify chain and add L hops



Add L hops

2.3 Blur (M, W)

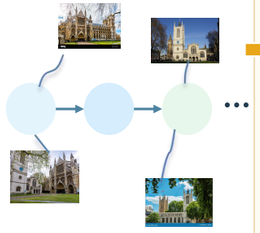
Resolve M nodes via indirect clues from W facts



M nodes x W cules

2.4 Visualize (N,P,Q)

Attach retrieved images and visual facts



N nodes x P images x Q facts

3 Question Synthesis

Rewrite into a harder question



In a district scene associated with a building neighboring the landmark shown in the image and now housing an aquarium in its basement, the pale Gothic tower on the right belongs to a church. In a photograph of an abbey associated with that church, how many arched doorways are visible at the base of the west front?

- Replace entity names with indirect clues
- Keep the original input image

4 Difficulty and Quality Control

Controllable difficulty and high-quality data

Difficulty Stratification



Control knobs

L M W
N P Q

Quality Control

- Answer uniqueness
- Evidence necessity on intended chain
- Verified web / image evidence
- Empirical shortcut checks

B Agentic Training: *Joint RL + On-Policy Distillation on Student-Generated Search Trajectories*

5 Trajectory Generation

Long-horizon Search Environment

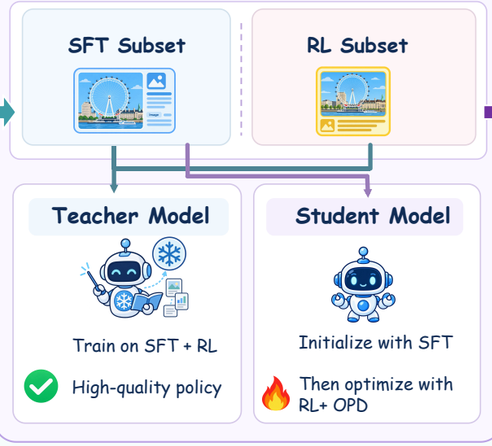


Successful supervision trajectories



6 Difficulty Split & Supervised

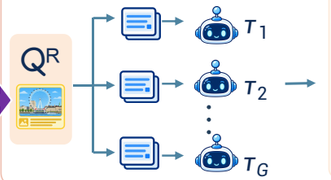
Use both SFT and RL data for teacher, and initialize student with SFT



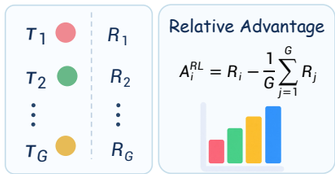
7 Joint RL + OPD

7.1 Rollout & Evaluation

Sample G student rollouts



7.2 GRPO



7.3 Reward-Conditioned OPD

(a) Gate
$g_i = \mathbf{1}(c_i < 1)$

Teacher scores are assigned only to failed or imperfect trajectories.

(b) Decay λ_k

Exponential decay over steps

(c) Token feedback (OPD)

$A_{i,t}^{OPD} = g_i \cdot sg(\omega_i \cdot \Delta_{i,t})$

Low-reward trajectories get stronger guidance

7.4 Joint Optimization

Combine RL and OPD signals

$$1 = 1^{RL} + \lambda \cdot 1^{OPD} + \beta 1^{KL}$$

Update with joint loss

8 Outcome

Improved long-horizon multimodal search capability

MMSearch-OPD Agent



Explore. Verify. Reason. Discover.



More capable, reliable, and general multimodal search agents